



Product Specification

M240UAN02.2

AU OPTRONICS CORPORATION

() Preliminary Specification
(V) Final Specification

Module	24" Color TFT-LCD
Model Name	M240UAN02.2

<table><tr><td>Customer</td><td>Date</td></tr><tr><td>_____</td><td>_____</td></tr><tr><td>Approved by</td><td></td></tr><tr><td>_____</td><td>_____</td></tr></table>	Customer	Date	_____	_____	Approved by		_____	_____	<table><tr><td>Approved by</td><td>Date</td></tr><tr><td><u>Howard Lee</u></td><td><u>Apr 30, 2018</u></td></tr><tr><td>Prepared by</td><td>Date</td></tr><tr><td><u>Louis Jung</u></td><td><u>Apr 30, 2018</u></td></tr></table>	Approved by	Date	<u>Howard Lee</u>	<u>Apr 30, 2018</u>	Prepared by	Date	<u>Louis Jung</u>	<u>Apr 30, 2018</u>
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Note: This Specification is subject to change without notice.	AU Optronics corporation																

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Record of Revision

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T _{disp} (h) ⁴⁰⁾	Active ⁴¹⁾	960 ⁴²⁾	960 ⁴³⁾	960 ⁴⁴⁾	Tclk ⁴⁵⁾																																																																																																																																																										
T _{blk} (h) ⁴⁶⁾	Blanking ⁴⁷⁾	40 ⁴⁸⁾	80 ⁴⁹⁾	468 ⁵⁰⁾	Tclk ⁵¹⁾																																																																																																																																																										
F _h ⁵²⁾	Frequency ⁵³⁾	59.5 ⁵⁴⁾	74 ⁵⁵⁾	85 ⁵⁶⁾	KHz ⁵⁷⁾	Note 3-3 ⁵⁸⁾																																																																																																																																																									
T _{clk} ⁵⁹⁾	Period ⁶⁰⁾	11.8 ⁶¹⁾	13 ⁶²⁾	16.8 ⁶³⁾	ns ⁶⁴⁾	1/fclk ⁶⁵⁾																																																																																																																																																									
fclk ⁶⁶⁾	Frequency ⁶⁷⁾	88.2 ⁶⁸⁾	77 ⁶⁹⁾	85 ⁷⁰⁾	MHz ⁷¹⁾	Note 3-4 ⁷²⁾																																																																																																																																																									

1 Handling Precautions

- 1) Since polarizer is easily damaged, do not touch or press the surface of polarizer with hand.
- 2) Be sure to turn off power supply when inserting or disconnecting from input connector.
- 3) Wipe off water drop immediately. Long contact with water may cause discoloration or spots.
- 4) When the panel surface is soiled, wipe it with absorbent cotton or other soft cloth.
- 5) Since the panel is made of glass, it may break or crack if dropped or bumped on hard surface.
- 6) Since CMOS LSI is used in this module, take care of static electricity and insure human earth when handling.
- 7) Do not open or modify the Module Assembly.
- 8) Do not press the reflector sheet at the back of the module to any directions.
- 9) In case a TFT-LCD Module has to be put back into the packing container slot after once it was taken out from the container, do not press the center of the LED lightbar edge. Otherwise the TFT-LCD Module may be damaged.
- 10) Insert or pull out the interface connector, be sure not to rotate nor tilt it of the TFT-LCD Module.
- 11) Do not twist nor bend the TFT -LCD Module even momentary. It should be taken into consideration that no bending/twisting forces are applied to the TFT-LCD Module from outside. Otherwise the TFT-LCD Module may be damaged.
- 12) Please avoid touching COF position while you are doing mechanical design.
- 13) When storing modules as spares for a long time, the following precaution is necessary: Store them in a dark place. Do not expose the module to sunlight or fluorescent light. Keep the temperature between 5°C and 35°C at normal humidity.
- 14) Do not apply the same pattern for a long time, it will enhance relevant defect.

2 General Description

This specification applies to the 24 inch wide Color a-Si TFT-LCD Module M240UAN02.2. The display supports the WUXGA - 1920(H) x 1200(V) screen format and 16.7M colors (8 bits RGB data input). The input interface is Dual channel LVDS and this module doesn't contain an driver board for backlight.

2.1 Display Characteristics

The following items are characteristics summary on the table under 25°C condition:

ITEMS	Unit	SPECIFICATIONS
Screen Diagonal	[mm]	611.32 (24.07")
Active Area	[mm]	518.4 (H) x 324.0 (V)
Pixels H x V	-	1920(x3) x 1200
Pixel Pitch	[um]	270 (per one triad) x270
Pixel Arrangement	-	R.G.B. Vertical Stripe
Display Mode	-	AHVA Mode (Advanced Hyper-Viewing Angle), Normally Black
EDR OFF White Luminance (Center)	[cd/m ²]	350 (Typ.)
EDR ON White Luminance (Center)	[cd/m ²]	400 (Typ.)
Contrast Ratio	-	1000 (Typ.)
Color Gamut	-	sRGB 99%
Response Time	[msec]	14 (Typ., GTG)
Power Consumption (LCD Module + Backligh unit)	[Watt]	17.6 @ EDR on 15.8 @ EDR off LCD module : PDD (Typ.)=4.5 @ White pattern,Fv=60 Hz Backlight unit : P _{BLU} (Typ.) =13.1 @EDR on Is=75mA Backlight unit : P _{BLU} (Typ.) =11.3 @EDR off Is=65mA
Weight	[Grams]	2530
Outline Dimension	[mm]	528.2(H) × 342.4(V) × 12.5(D) Typ.
Electrical Interface	-	Dual channel LVDS
Support Color	-	16.7M colors (RGB 8- bits)
Surface Treatment	-	Anti-Glare, SAG25
Temperature Range Operating Storage (Shipping)	[°C] [°C]	0 to +50 -20 to +60
RoHS Compliance	-	RoHS Compliance
TCO Compliance	-	TCO 7.0 Compliance



Product Specification

AU OPTRONICS CORPORATION

M240UAN02.2

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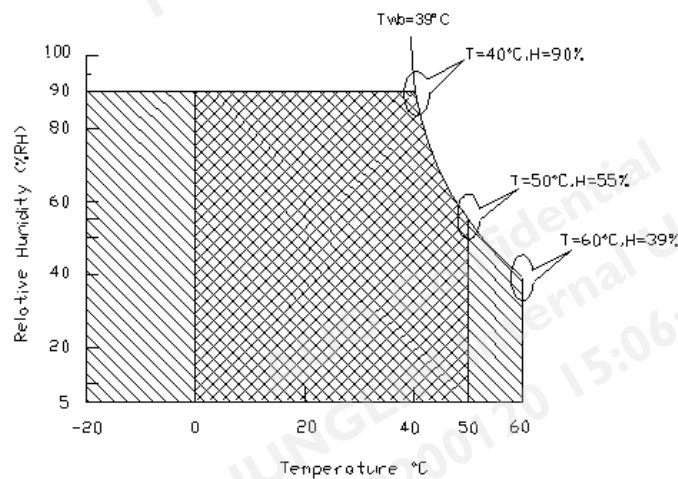
2.2 Absolute Maximum Rating of Environment

Permanent damage may occur if exceeding the following maximum rating.

Symbol	Description	Min.	Max.	Unit	Remark
TOP	Operating Temperature	0	+50	[°C]	Note 2-1
TGS	Glass surface temperature (operation)	0	+65	[°C]	Note 2-1 Function judged only
HOP	Operation Humidity	5	90	[%RH]	Note 2-1
TST	Storage Temperature	-20	+60	[°C]	
HST	Storage Humidity	5	90	[%RH]	

Note 2-1: Temperature and relative humidity range are shown as the below figure.

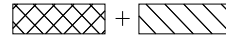
1. 90% RH Max ($T_a \leq 39^\circ\text{C}$)
2. Max wet-bulb temperature at 39°C or less. ($T_a \leq 39^\circ\text{C}$)
3. No condensation



Operating Range



Storage Range



2.3 Optical Characteristics

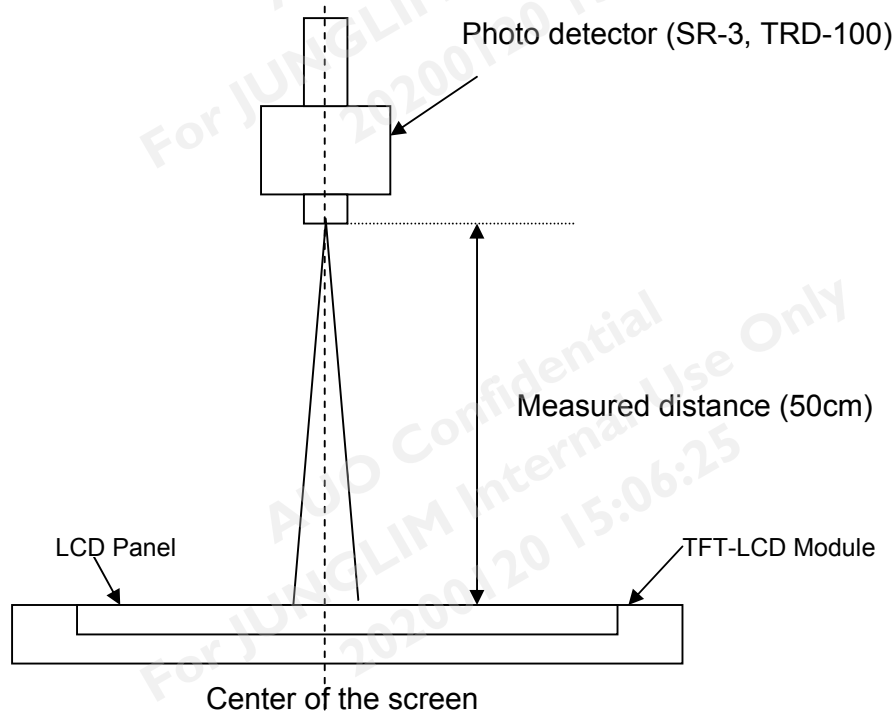
The optical characteristics are measured on the following test condition.

Test Condition:

1. Equipment setup: Please refer to **Note 2-2**.
2. Panel Lighting time: 30 minutes
3. VDD=5.0V, Fv=60Hz, Is=65mA, Ta=25°C

Symbol	Description		Min.	Typ.	Max.	Unit	Remark
L _w	White Luminance (Center of screen)		280	350	-	[cd/m2]	@EDR off Note 2-2 Note 2-3 By SR-3
EDR-L _w	White Luminance (10% of screen)		320	400	-	[cd/m2]	@EDR on Note 2-2 Note 2-4 By SR-3
L _{uni}	Luminance Uniformity (9 points)		75	80	-	[%]	Note 2-5 By SR-3
CR	Contrast Ratio (Center of screen)		600	1000	-	-	Note 2-6 By SR-3
θ _R	Horizontal Viewing Angle (CR=10)	Right	75	89	-	[degree]	Note 2-7 By SR-3
θ _L		Left	75	89	-		
Φ _H	Vertical Viewing Angle (CR=10)	Up	70	89	-		
Φ _L		Down	70	89	-		
θ _R	Horizontal Viewing Angle (CR=5)	Right	75	89	-		
θ _L		Left	75	89	-		
Φ _H	Vertical Viewing Angle (CR=5)	Up	70	89	-		
Φ _L		Down	70	89	-		
T _{GTG}	Response Time	Gray To Gray	-	14	-	[msec]	Note 2-8 By TRD-100
R _x	Color Coordinates (CIE 1931)	Red x	0.623	0.653	0.683	-	By SR-3
R _y		Red y	0.308	0.338	0.368		
G _x		Green x	0.278	0.308	0.338		
G _y		Green y	0.596	0.626	0.656		
B _x		Blue x	0.116	0.146	0.176		
B _y		Blue y	0.024	0.054	0.084		
W _x		White x	0.283	0.313	0.343		
W _y		White y	0.299	0.329	0.359		
sRGB coverage ratio				99		[%]	By SR-3

Note 2-2: Equipment setup :



Note 2-3: LED current condition @EDR off $I_s = 65\text{mA}$

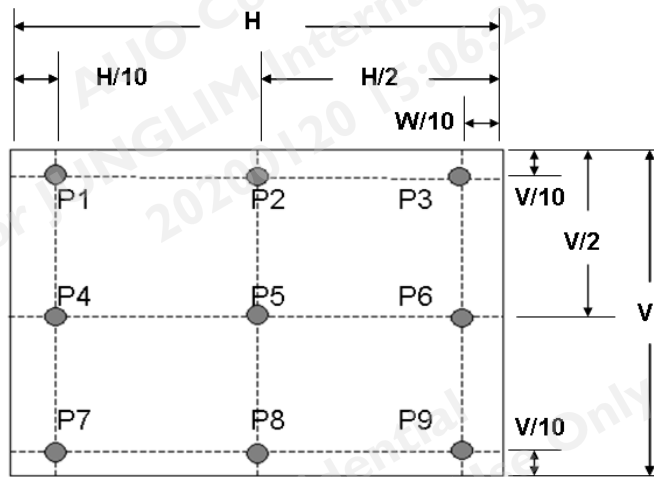
Note 2-4: LED current condition @EDR on $I_s = 75\text{mA}$

Note 2-5: Luminance Uniformity Measurement

Definition:

$$\text{Luminance Uniformity} = \frac{\text{Minimum Luminance of 9 Points (P1 ~ P9)}}{\text{Maximum Luminance of 9 Points (P1 ~ P9)}}$$

a. Test pattern: White Pattern



Note 2-6: Contrast Ratio Measurement

Definition:

$$\text{Contrast Ratio} = \frac{\text{Luminance of White pattern}}{\text{Luminance of Black pattern}}$$

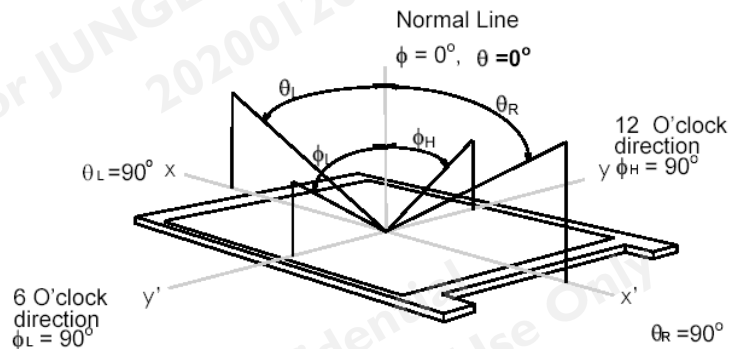
a. Measured position: Center of screen (P5) & perpendicular to the screen
($\theta = \phi = 0^\circ$)

Note 2-7: Viewing angle measurement

Definition: The angle at which the contrast ratio is greater than 10 & 5 .

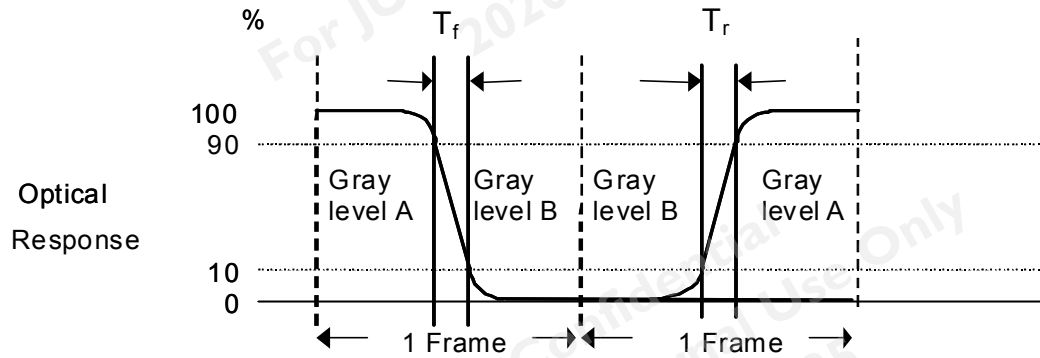
a. Horizontal view angle: Divide to left & right (θ_L & θ_R)

Vertical view angle: Divide to up & down (ϕ_H & ϕ_L)



Note 2-8: Response time measurement

The output signals of photo detector are measured when the input signals are changed from “Gray level A” to “Gray level B” (falling time, T_f), and from “Gray level B” to “Gray level A” (rising time, T_r), respectively. The response time is interval between the 10% and 90% of optical response.



The gray to gray response time is defined as the following table.

Gray Level to Gray Level		Target gray level				
		L0	L63	L127	L191	L255
Start gray level	L0					
	L63					
	L127					
	L191					
	L255					

■ T_{GTG_typ} is the total average time at rising time and falling time of gray to gray.

2.4 Mechanical Characteristics

Symbol	Description	Min.	Max.	Unit	Remark
P _{bc}	Backside Compression	2.5	-	[Kgf]	Note 2-9

Note 2-9: 1. Test condition:

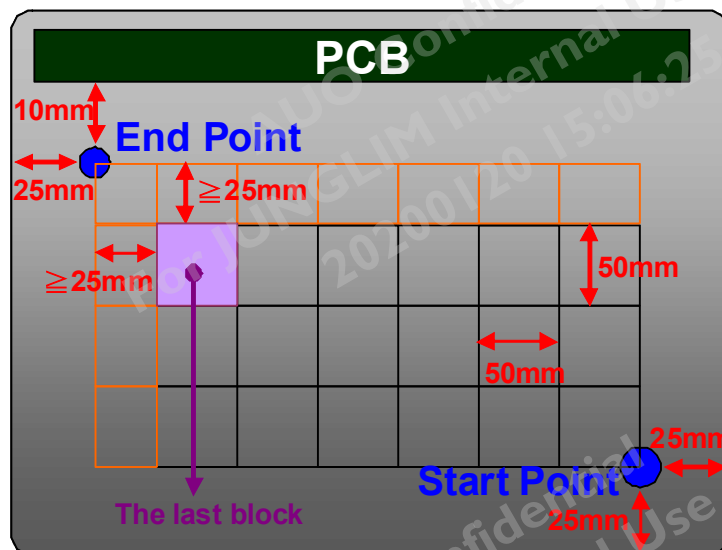
- (1) Ambient illumination : 10 ~ 15 lux
- (2) Test pattern : Gray (L128)
- (3) Distance of judgment : 30 cm from surface of LCD panel
- (4) Viewing angle of judgment : Front
- (5) Test jig : Push-Pull Gauge & Flat Head (Φ2)

2. Test Method:

The point is at a distance from right-downside 25mm x 25mm defined as the Start Point of Measure Points, and the point is at a distance 25mm from left-side & around 10mm from PCB defined as the End Point.

Align 50mm x 50mm block from Start Point on the Bezel Back, and the corners of each block are Measure Points.

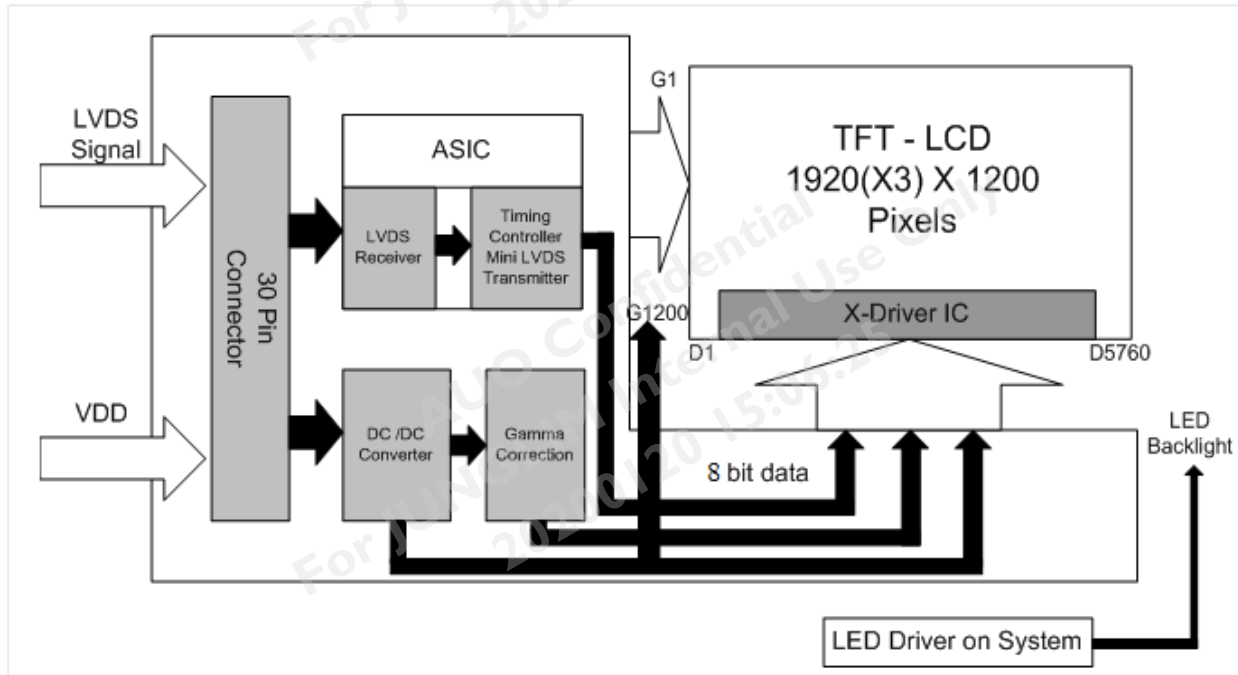
If the distance from the last block to each side of the End Point ≥ 25 mm, add other blocks to make sure that most area of Bezel Back can be measured.



3 TFT-LCD Module

3.1 Block Diagram

The following shows the block diagram of the 24 inch Color TFT-LCD Module.



3.2 Interface Connection

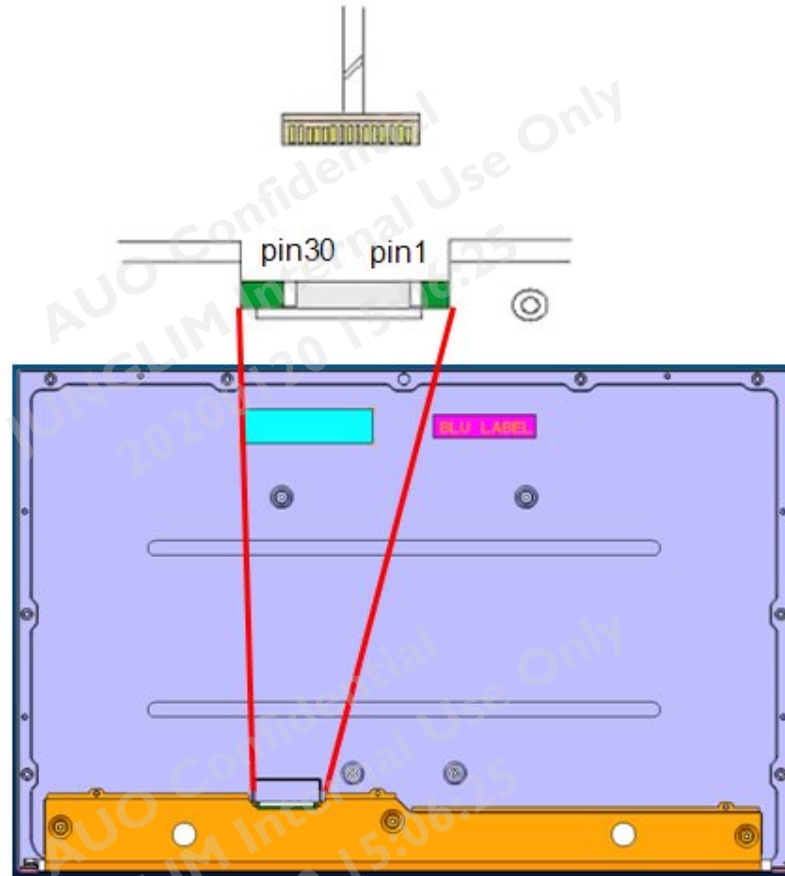
3.2.1 Connector Type

TFT-LCD Connector	Manufacturer	P-TWO	STM
	Part Number	187034-3009	MSBKT2407P30HB
Mating Connector	Manufacturer	JAE or Compatible	
	Part Number	FI-X30HL (Locked Type)	

3.2.2 Connector Pin Assignment

PIN #	Symbol	Description	Remark
1	RxO0-	Negative LVDS differential data input (Odd data)	
2	RxO0+	Positive LVDS differential data input (Odd data)	
3	RxO1-	Negative LVDS differential data input (Odd data)	
4	RxO1+	Positive LVDS differential data input (Odd data)	
5	RxO2-	Negative LVDS differential data input (Odd data)	
6	RxO2+	Positive LVDS differential data input (Odd data)	
7	GND	Ground	
8	RxOCLK-	Negative LVDS differential clock input (Odd clock)	
9	RxOCLK+	Positive LVDS differential clock input (Odd clock)	
10	RxO3-	Negative LVDS differential data input (Odd data)	
11	RxO3+	Positive LVDS differential data input (Odd data)	
12	RxE0-	Negative LVDS differential data input (Even data)	
13	RxE0+	Positive LVDS differential data input (Even data)	
14	GND	Ground	
15	RxE1-	Negative LVDS differential data input (Even data)	
16	RxE1+	Positive LVDS differential data input (Even data)	
17	GND	Ground	
18	RxE2-	Negative LVDS differential data input (Even data)	
19	RxE2+	Positive LVDS differential data input (Even data)	
20	RxECLK-	Negative LVDS differential clock input (Even clock)	
21	RxECLK+	Positive LVDS differential clock input (Even clock)	
22	RxE3-	Negative LVDS differential data input (Even data)	
23	RxE3+	Positive LVDS differential data input (Even data)	
24	GND	Ground	
25	NC	No connection (for AUO test only. Do not connect)	
26	NC	No connection (for AUO test only. Do not connect)	

27	NC	No connection (for AUO test only. Do not connect)	
28	VDD	Power Supply Input Voltage	
29	VDD	Power Supply Input Voltage	
30	VDD	Power Supply Input Voltage	



3.3 Electrical Characteristics

3.3.1 Absolute Maximum Rating

Permanent damage may occur if exceeding the following maximum rating.

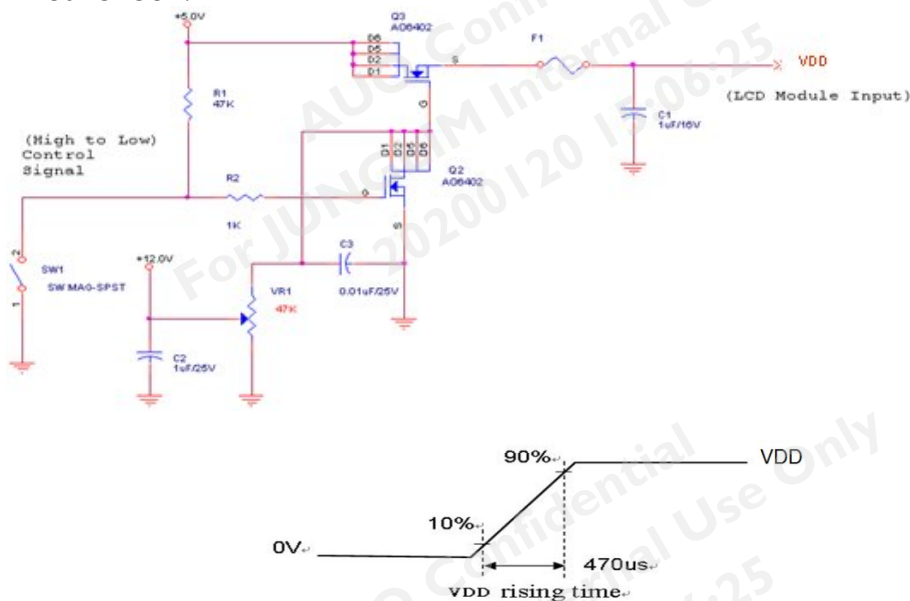
Symbol	Description	Min	Max	Unit	Remark
VDD	Power Supply Input Voltage	GND-0.3	6.0	[Volt]	Ta=25°C

3.3.2 Recommended Operating Condition

Symbol	Description	Min	Typ	Max	Unit	Remark
VDD	Power supply Input voltage	4.5	5	5.5	[Volt]	
IDD	Power supply Input Current (RMS)	-	0.90	1.08	[A]	VDD= 5.0V, Whie Pattern, Fv=60Hz
			1.03	1.24	[A]	VDD= 5.0V, Whie Pattern, Fv=70Hz
PDD	VDD Power Consumption	-	4.50	5.40	[Watt]	VDD= 5.0V, Whie Pattern, Fv=60Hz
			5.15	6.20	[Watt]	VDD= 5.0V, Whie Pattern, Fv=70Hz
IRush	Inrush Current	-	-	3	[A]	Note 3-1
VDDrp	Allowable VDD Ripple Voltage	-	-	500	[mV]	VDD= 5.0V, White Pattern, Fv=70Hz

Note 3-1: Inrush Current measurement:

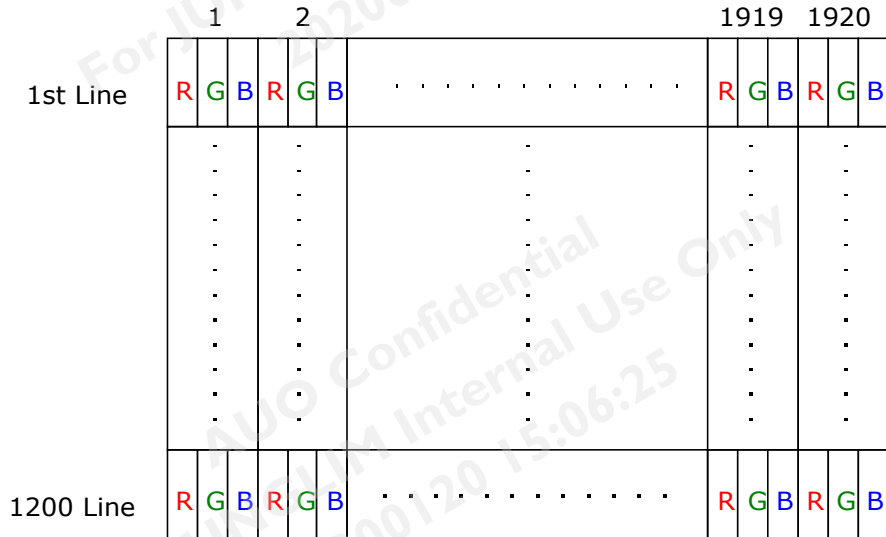
Test circuit:



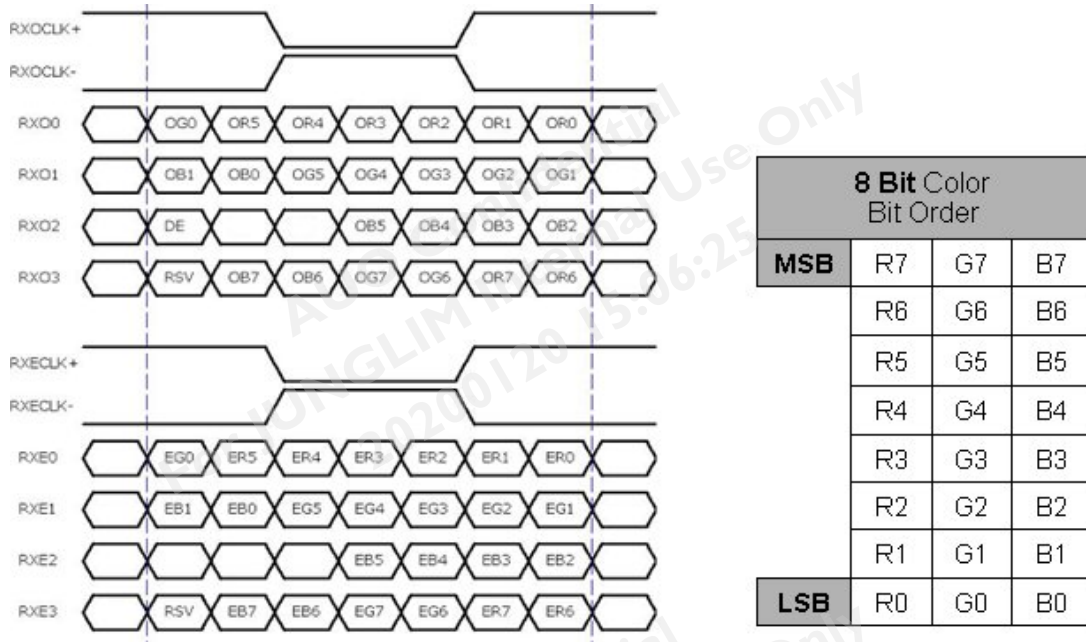
The duration of VDD rising time: 470us.

3.4 Signal Characteristics

3.4.1 LCD Pixel Format



3.4.2 LVDS Data Format



Note 3-2:

- O = "Odd Pixel Data" E = "Even Pixel Data"
- Refer to 3.4.1 LCD pixel format, the 1st data is 1 (Odd Pixel Data), the 2nd data is 2 (Even Pixel Data) and the last data is 1920 (Even Pixel Data).

3.4.3 Color versus Input Data

The following table is for color versus input data (8bit). The higher the gray level, the brighter the color.

Color	Gray Level	Color Input Data																								Remark
		RED data (MSB :R7, LSB :R0)								GREEN data (MSB :G7, LSB :G0)								BLUE data (MSB :B7, LSB :B0)								
		R7	R6	R5	R4	R3	R2	R1	R0	G7	G6	G5	G4	G3	G2	G1	G0	B7	B6	B5	B4	B3	B2	B1	B0	
Black	-	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
White	-	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	
Gray 127	-	0	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	
Red	L0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Black
	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	
	L255	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Green	L0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Black
	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	
	L255	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	
Blue	L0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Black
	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	
	L255	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	

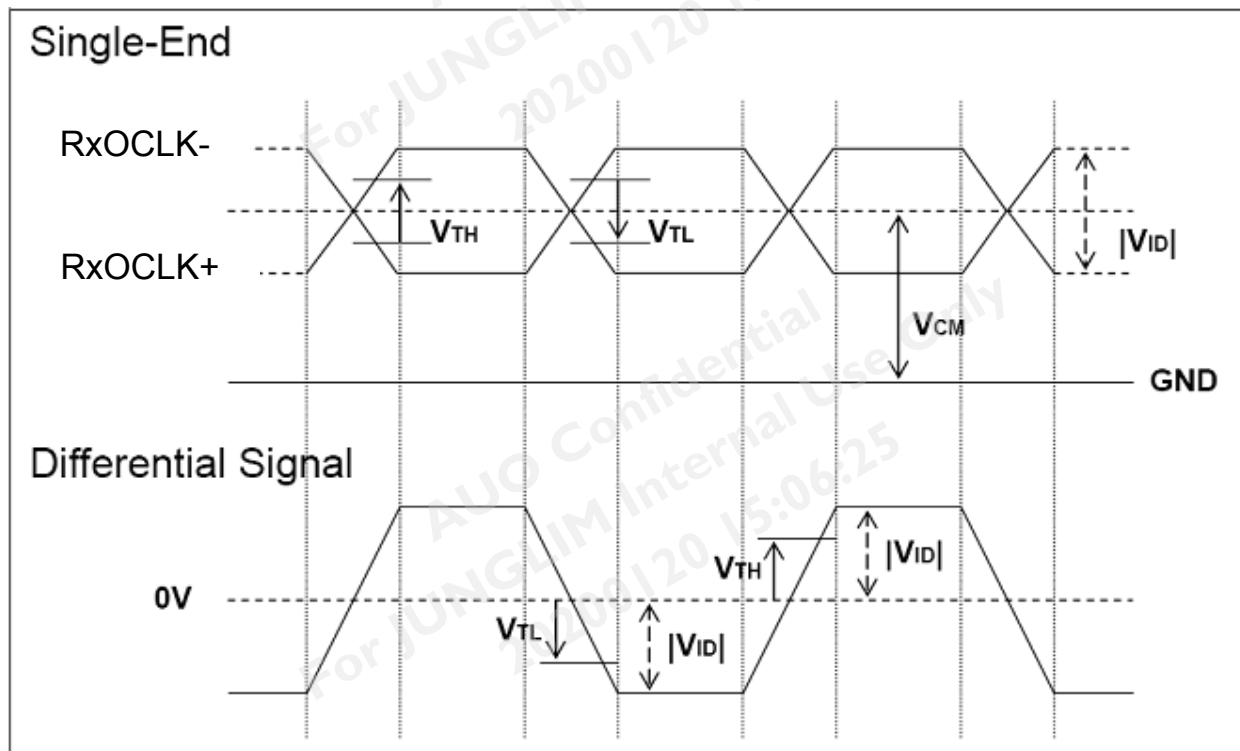
3.4.4 LVDS Specification

a. DC Characteristics:

Symbol	Description	Min	Typ	Max	Units	Condition
V_{TH}	LVDS Differential Input High Threshold	-	-	+100	[mV]	$V_{CM} = 1.2V$
V_{TL}	LVDS Differential Input Low Threshold	-100	-	-	[mV]	$V_{CM} = 1.2V$
$ V_{ID} $	LVDS Differential Input Voltage	100	-	600	[mV]	
V_{CM}	LVDS Common Mode Voltage	+1.0	+1.2	+1.5	[V]	$V_{TH} - V_{TL} = 200mV$

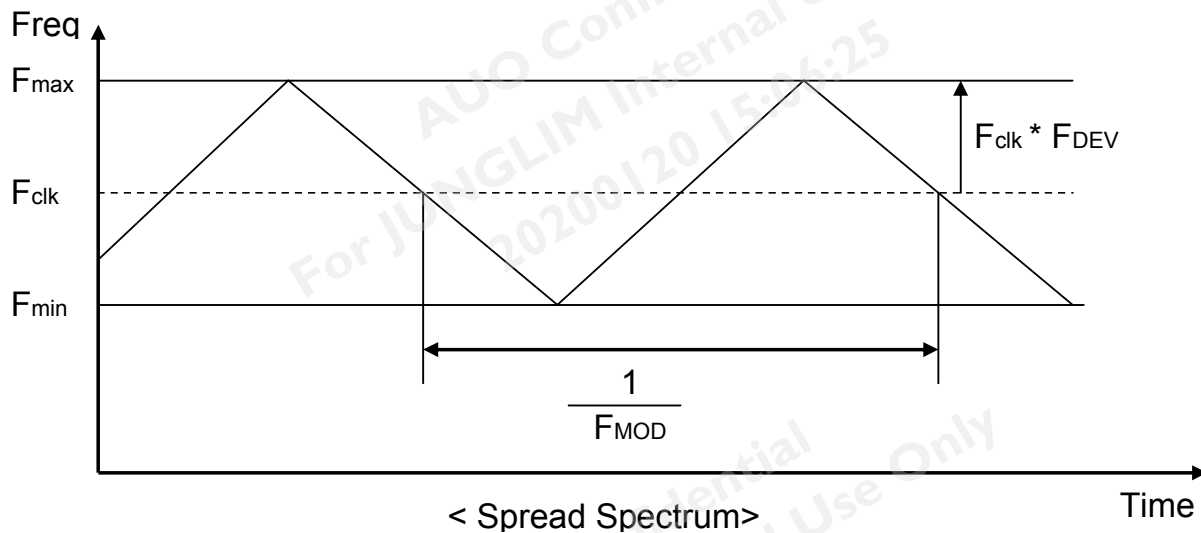
LVDS Signal Waveform:

Use RxOCLK- & RxOCLK+ as example.



b. AC Characteristics:

Symbol	Description	Min	Max	Unit	Remark
F_{DEV}	Maximum deviation of input clock frequency during Spread Spectrum	-	± 3	%	
F_{MOD}	Maximum modulation frequency of input clock during Spread Spectrum	-	200	KHz	



F_{clk} : LVDS Clock Frequency

3.4.5 Input Timing Specification

It only support DE mode, and the input timing are shown as the following table.

Symbol	Description		Min.	Typ.	Max.	Unit	Remark
Tv	Vertical Section	Period	1214	1235	1734	Th	
Tdisp (v)		Active	1200	1200	1200	Th	
Tblk (v)		Blanking	14	35	534	Th	
Fv		Frequency	48	60	70	Hz	
Th	Horizontal Section	Period	1000	1040	1428	Tclk	
Tdisp (h)		Active	960	960	960	Tclk	
Tblk (h)		Blanking	40	80	468	Tclk	
Fh		Frequency	59.5	74	85	KHz	Note 3-3
Tclk	LVDS Clock	Period	11.8	13	16.8	ns	1/Fclk
Fclk		Frequency	58.2	77	85	MHz	Note 3-4

Note 3-3: The equation is listed as following. Please don't exceed the above recommended value.

$$Fh (\text{Min.}) = Fclk (\text{Min.}) / Th (\text{Min.});$$

$$Fh (\text{Typ.}) = Fclk (\text{Typ.}) / Th (\text{Typ.});$$

$$Fh (\text{Max.}) = Fclk (\text{Max.}) / Th (\text{Min.});$$

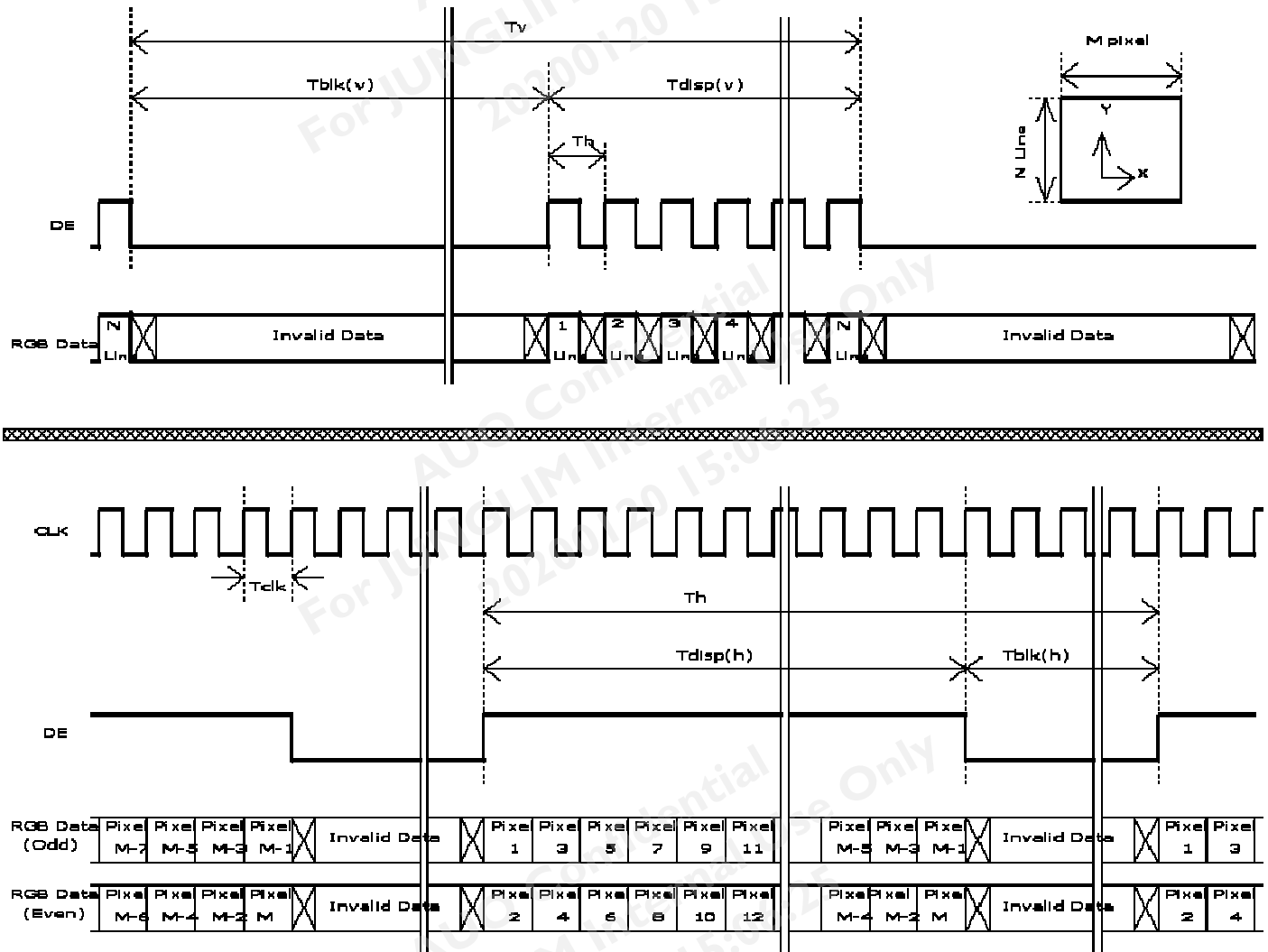
Note 3-4: The equation is listed as following. Please don't exceed the above recommended value.

$$Fclk (\text{Min.}) = Fv (\text{Min.}) \times Th (\text{Min.}) \times Tv (\text{Min.});$$

$$Fclk (\text{Typ.}) = Fv (\text{Typ.}) \times Th (\text{Typ.}) \times Tv (\text{Typ.});$$

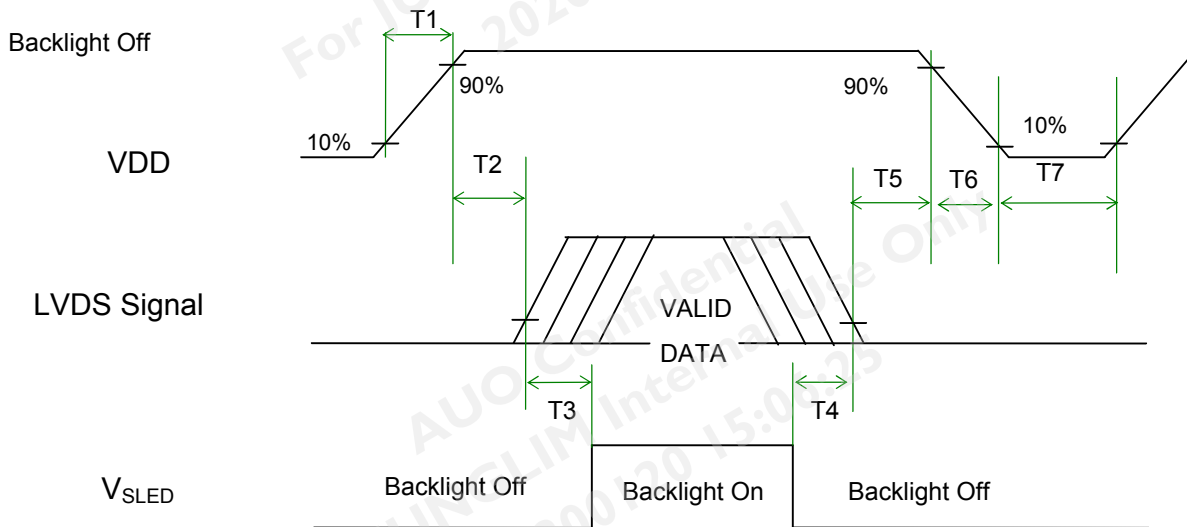
$$Fclk (\text{Max.}) = Fv (\text{Max.}) \times Th (\text{Typ.}) \times Tv (\text{Typ.});$$

3.4.6 Input Timing Diagram



3.5 Power ON/OFF Sequence

VDD power, LVDS signal and backlight on/off sequence are as following. LVDS signals from any system shall be Hi-Z state when VDD is off.



Power Sequence Timing

Symbol	Value			Unit	Remark
	Min.	Typ.	Max.		
T1	0.5	-	10	[ms]	
T2	0	-	50	[ms]	
T3	500	-	-	[ms]	
T4	100	-	-	[ms]	
T5	0	-	50	[ms]	Note 3-5 Note 3-6
T6	0	-	200	[ms]	Note 3-6 Note 3-7
T7	1000	-	-	[ms]	

Note 3-5 : Recommend setting T5 = 0ms to avoid electronic noise when VDD is off.

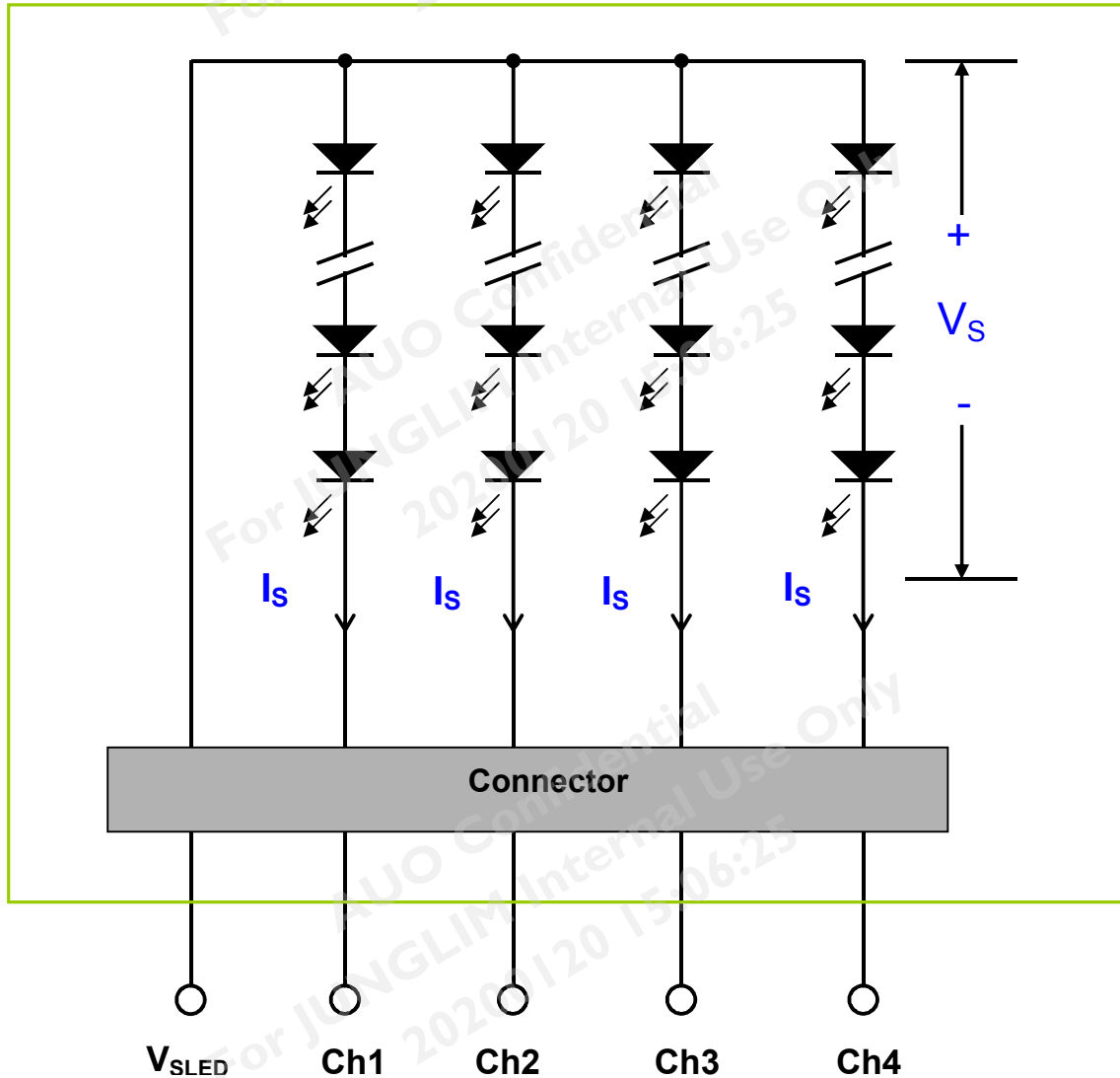
Note 3-6 : During T5 and T6 period , please keep the level of input LVDS signals with Hi-Z state.

Note 3-7 : Voltage of VDD must decay smoothly after power-off. (customer system decide this value)

4 Backlight Unit

4.1 Block Diagram

The following shows the block diagram of the 24 inch Backlight Unit. And it includes 60 pcs LED in the LED light bar. (4 strings and 15 pcs LED of one string).



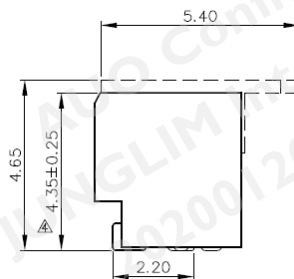
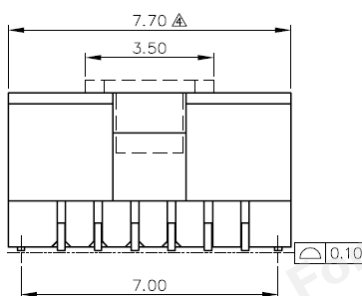
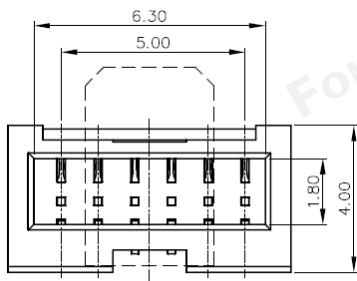
4.2 Interface Connection

4.2.1 Connector Type

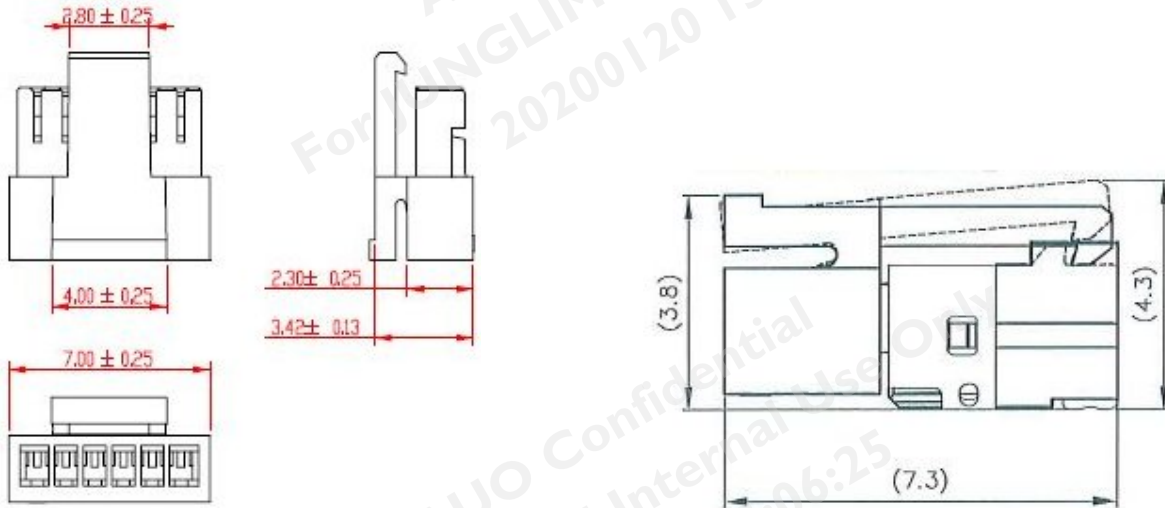
Backlight Connector	Manufacturer	ENTERY
	Part Number	3709K-Q06C-04L
Mating Connector	Manufacturer	ENTERY
	Part Number	H112K-P06N-00B (Non-Locking type) H112K-P06N-03B (Locking type)

Backlight Connector dimension:

$H \times V \times D = 7.7 \times 4.35 \times 4$, Pitch=1.0(unit=mm),use 6 pin connector

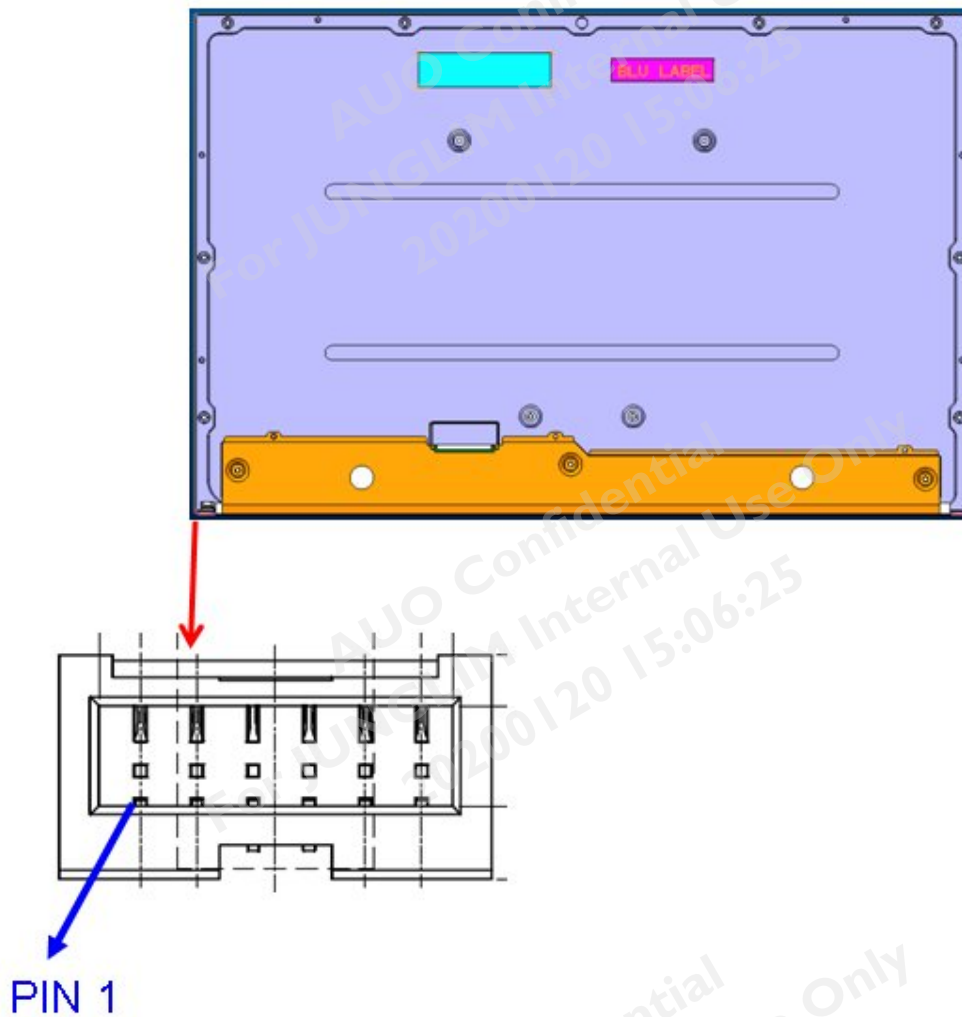


Mating Connector dimension:



4.2.2 Connector Pin Assignment

Pin#	Symbol	Description	Remark
1	Ch1	LED Current Feedback Terminal (Channel 1)	
2	Ch2	LED Current Feedback Terminal (Channel 2)	
3	V _{SLED}	LED Power Supply Voltage Input Terminal	
4	V _{SLED}	LED Power Supply Voltage Input Terminal	
5	Ch3	LED Current Feedback Terminal (Channel 3)	
6	Ch4	LED Current Feedback Terminal (Channel 4)	



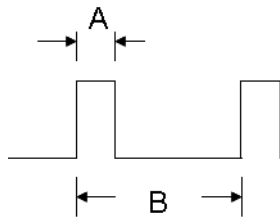
4.3 Electrical Characteristics

4.3.1 Absolute Maximum Rating

Permanent damage may occur if exceeding the following maximum rating.

(Ta=25°C)

Symbol	Description	Min	Max	Unit	Remark
Is	LED String Current	0	150	[mA]	100% duty ratio
			300	[mA]	Duty ratio ≤ 10% Pulse time=10 ms



Duty ratio= (A / B) X 100% ; (A: Pulse time, B: Period)

4.3.2 Recommended Operating Condition

(Ta=25°C)

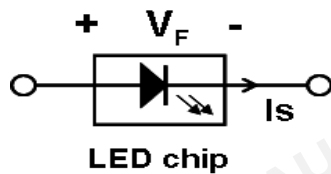
Symbol	Description	Min.	Typ.	Max.	Unit	Remark
Is	LED String Current	-	65	72	[mA]	@EDR off 100% duty ratio of LED chip Note 4-7
Is	LED String Current	-	75	83	[mA]	@EDR on 100% duty ratio of LED chip Note 4-7
Vs	LED String Voltage	39	43.5	48	[Volt]	Is=65mA @ 100% duty ratio; Note 4-1, Note 4-5
ΔVs	Maximum Vs Voltage Deviation of light bar	-	-	3	[Volt]	Is=65mA @ 100% duty ratio; Note 4-2
P _{BLU}	LED Light Bar Power Consumption	-	11.3	12.5	[Watt]	@EDR off Note 4-3
P _{BLU}	LED Light Bar Power Consumption	-	13.1	14.4	[Watt]	@EDR on Note 4-3

LT _{LED}	LED Life Time	30,000	-	-	[Hour]	Note 4-4
OVP	Over Voltage Protection in system board	110% V _{smax}	-	-	[Volt]	Note 4-5

Note 4-1: $V_s (\text{Typ.}) = V_F (\text{Typ.}) \times \text{LED No. (one string)}$;

a. V_F : LED chip forward voltage, $V_F (\text{Min.}) = 2.6\text{V}$, $V_F (\text{Typ.}) = 2.9\text{V}$, $V_F (\text{Max.}) = 3.2\text{V}$

b. The same equation to calculate $V_s (\text{Min.})$ & $V_s (\text{Max.})$ for respective $V_F (\text{Min.})$ & $V_F (\text{Max.})$;



Note 4-2: $\Delta V_s (\text{Max.}) = \Delta V_F \times \text{LED No. (one string)}$;

a. ΔV_F : LED chip forward voltage deviation; (0.2 V , each Bin of LED V_F)

Note 4-3: $P_{\text{BLU}} (\text{Typ.}) = V_s (\text{Typ.}) \times I_s (\text{Typ.}) \times 4$; (4 is total String No. of LED Light bar)

$P_{\text{BLU}} (\text{Max.}) = V_s (\text{Max.}) \times I_s (\text{Typ.}) \times 4$;

Note 4-4: Definition of life time:

a. Brightness of LED becomes to 50% of its original value

b. Test condition: $I_s = 65\text{mA}$ and 25°C (Room Temperature)

Note 4-5: Recommendation for LED driver power design:

Due to there are electrical property deviation in LED & monitor set system component after long time operation. AUO strongly recommend the design value of LED driver board OVP (over voltage protection) should be 10% higher than max. value of LED string voltage (V_s) at least.

Note 4-6: AUO strongly recommend "Analog Dimming" method for backlight brightness control for Wavy Noise Free. Otherwise, recommend that Dimming Control Signal (PWM Signal) should be synchronized with Frame Frequency.

Note 4-7 Ensure that the LED light bar is not subjected either forward or reverse voltage while monitor set is on standby mode or not in use.



Product Specification

AU OPTRONICS CORPORATION

M240UAN02.2

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5 Reliability Test

AUO reliability test items are listed as following table. (Bare Panel only)

Items	Condition	Remark
Temperature Humidity Bias (THB)	Ta= 50°C , 80%RH, 300hours	
High Temperature Operation (HTO)	Ta= 50°C , 50%RH, 300hours	
Low Temperature Operation (LTO)	Ta= 0°C , 300hours	
High Temperature Storage (HTS)	Ta= 60°C , 300hours	
Low Temperature Storage (LTS)	Ta= -20°C , 300hours	
Vibration Test (Non-operation)	Acceleration: 1.5 Grms Wave: Random Frequency: 10 - 200 Hz Sweep: 30 Minutes each Axis (X, Y, Z)	
Shock Test (Non-operation)	Acceleration: 50 G Wave: Half-sine Active Time: 20 ms Direction: ±X, ±Y, ±Z (one time for each Axis)	
Thermal Shock Test (TST)	-20°C /30min, 60°C /30min, 100 cycles	Note 5-1
On/Off Test	On/10sec, Off/10sec, 30,000 cycles	
ESD (Electro Static Discharge)	Contact Discharge: ± 15KV, 150pF(330Ω) 1sec, 8 points, 25 times/ point.	Note 5-2
	Air Discharge: ± 15KV, 150pF(330Ω) 1sec 8 points, 25 times/ point.	
Altitude Test	Operation:18,000 ft Non-Operation:40,000 ft	

Note 5-1: a. A cycle of rapid temperature change consists of varying the temperature from -20°C to 60°C, and back again. Power is not applied during the test.
b. After finish temperature cycling, the unit is placed in normal room ambient for at least 4 hours before power on.

Note 5-2: EN61000-4-2, ESD class B: Certain performance degradation allowed
No data lost
Self-recoverable
No hardware failures.

ESD discharged points should avoid display area and periphery front bezel of display area. Suggest points were 4 side parallel edge of display area surface. Metal front bezel must cover half area of BM (black matrix), and metal front bezel must connect with metal back bezel to protect source IC of panel by ESD

damaged.

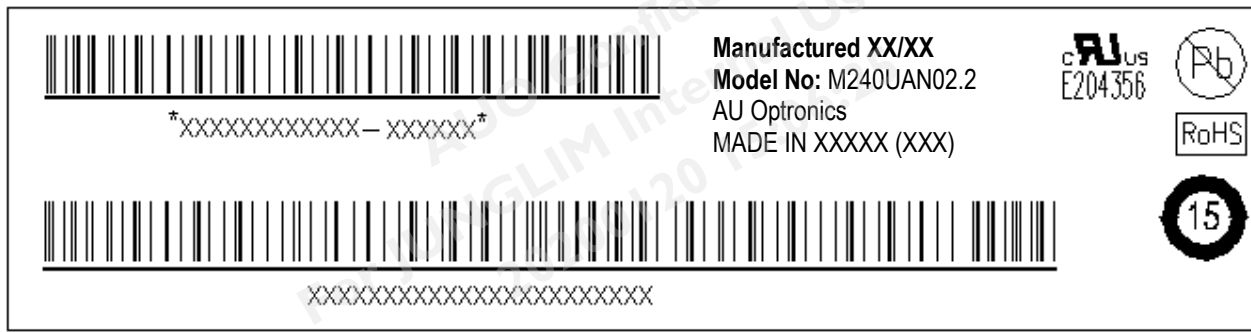
Note 5-3: Result Evaluation Criteria:

TFT-LCD panels test should take place after gradually cooling enough at room temperature.

In the normal application, there should be no particular problems that may affect the display function.

6 Shipping Label

The label is on the panel as shown below:



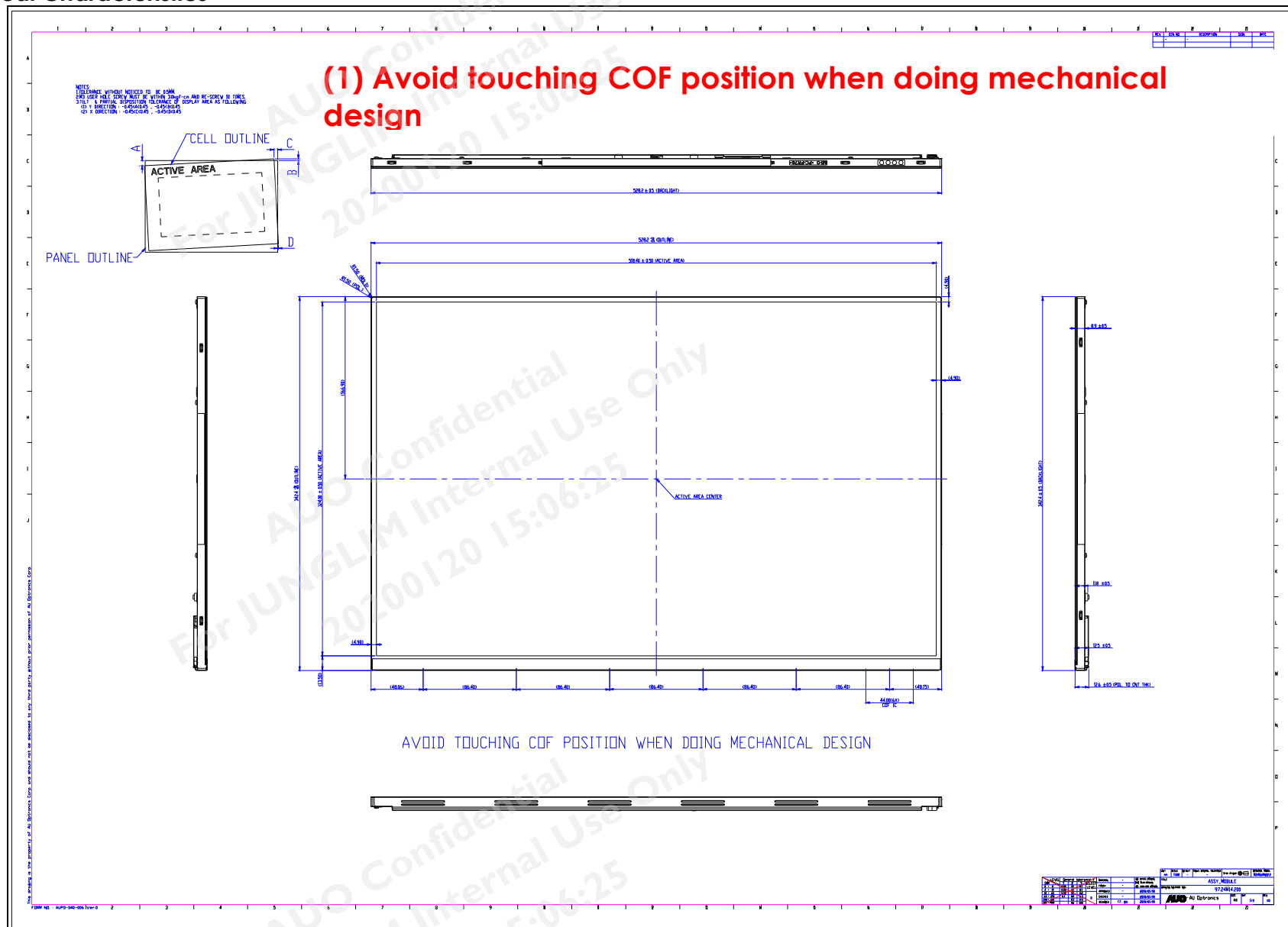
Note 6-1: For Pb Free products, AUO will add  for identification.

Note 6-2: For RoHS compatible products, AUO will add  for identification.

Note 6-3: For China RoHS compatible products, AUO will add  for identification.

Note 6-4: The Green Mark will be presented only when the green documents have been ready by AUO Internal Green Team.

7 Mechanical Characteristics

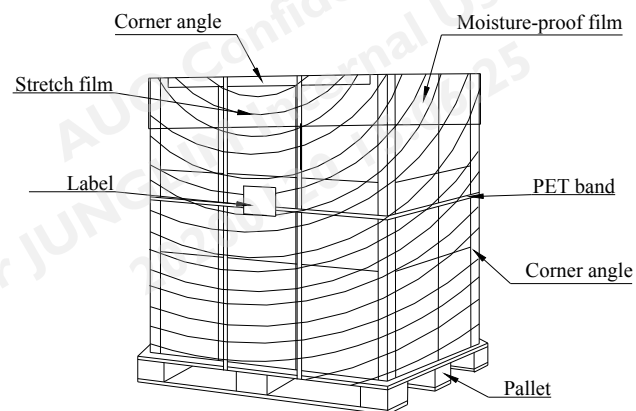
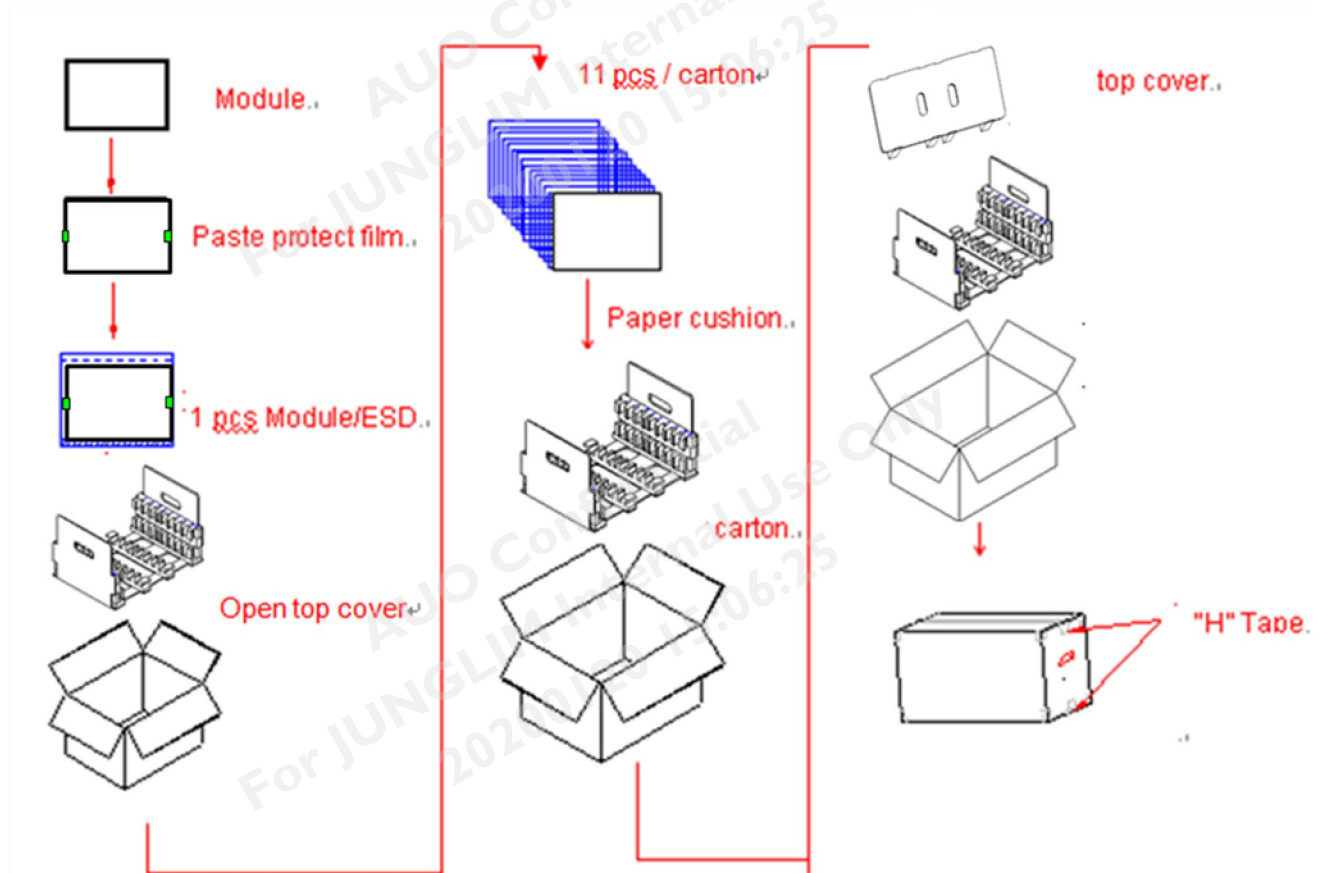


Ver 1.1



8 Packing Specification

8.1 Packing Flow



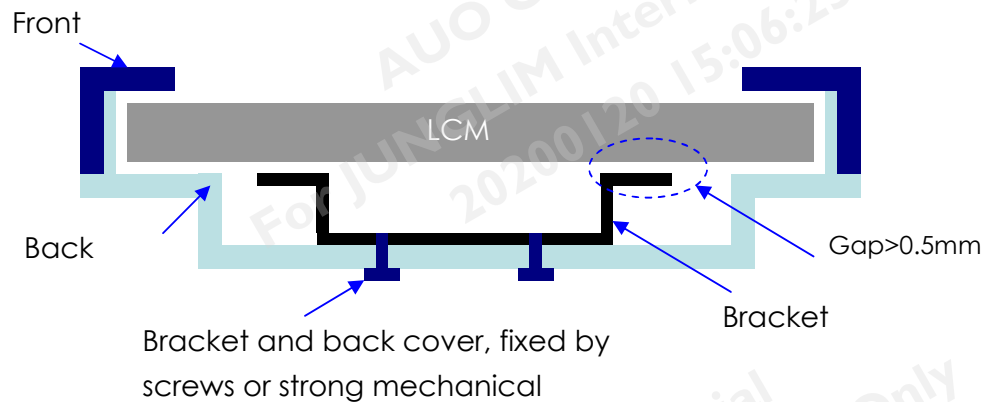
8.2 Pallet and shipment information

Item	Specification			Remark
	Q'ty	Dimension	Weight(kg)	
Panel	1	528.2(H) × 342.4(V) × 12.5(D) Typ..	2.53	
Cushion	1	-	0.6	
Box	1	619(L)mm x 373(W)mm x427(H)mm	1.5	without Panel & cushion
Packing Box	11 pcs/Box	619(L)mm x 373(W)mm x427(H)mm	29.93	with panel & cushion
Pallet	1	1315(L)mm x1150(W)mm x132(H)mm	17.00	
Pallet after Packing	18 boxes/pallet	1315(L)mm x1150(W)mm x132(H)mm	555.74	

9 Design Guide for System

9.1 The gap between LCM and system rear bracket should be bigger than 0.5mm.

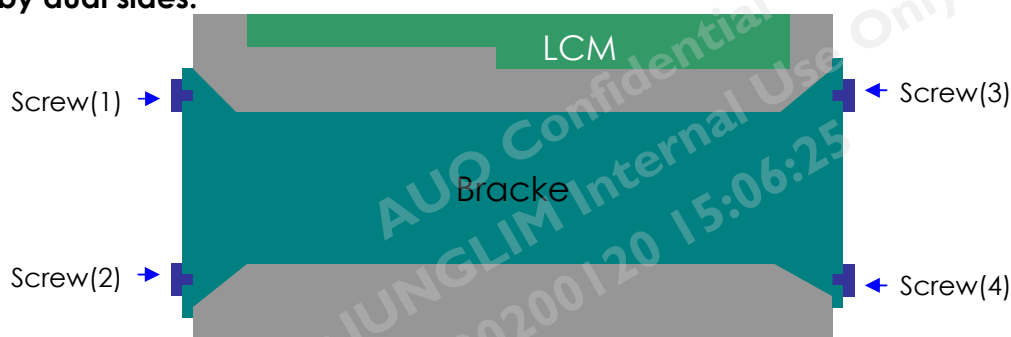
9.2 The system bracket should be fixed on back cover firmly.



9.3 The EMI gasket should be uniform and not push panel strongly.



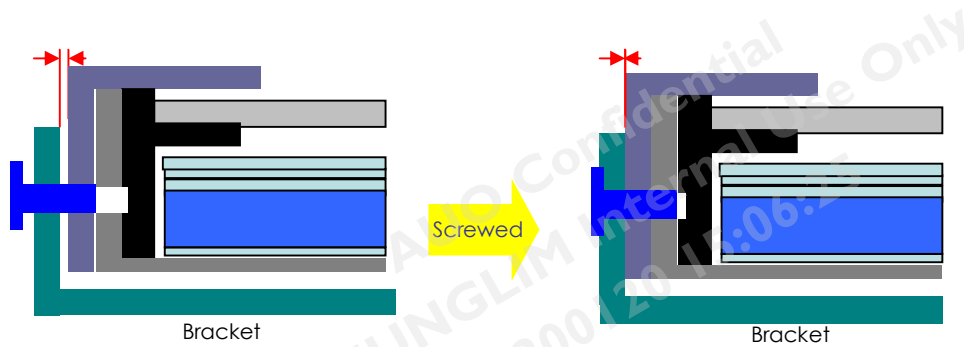
9.4 For stable assembly, the system bracket should use 4 screws to fix system and panel by dual sides.



9.5 The system bracket and panel should be in parallel with having no gap after inserting screws.

Proper and Parallel

0 gap and no mechanical damage



9.6 Avoid scratching LCM, the rib on system front-cover should not exceed the bottom edge of LCM's front-bezel.

